

BRAC University
Department of Electrical and Electronic Engineering
EEE/ECE-103IL: Computer Programming Laboratory
Summer 2026 (Section: 04)

Lab Final

1. Develop a **C program** that dynamically allocates memory to store the **weights of N packages** (in kg) to be loaded onto a cargo truck. The value of **N** and all the values will be entered by the user. The weights are recorded in the order the packages are received, which follows no specific pattern. [CO4] (10)

To keep the truck balanced, the loading plan must report:

- the weight of the **2nd heaviest** package (it is placed over the rear axle)
- the weight of the **4th lightest** package (it is placed near the door)
- how much heavier the 2nd heaviest is compared to the 4th lightest one

Write a C program that performs the following tasks:

- Dynamically allocate memory to store the N values
- Read the N values from the user
- Determine the values asked for above
- Display the results with clear labels

Constraints:

- Dynamic memory allocation is mandatory
- Do not use any built-in functions
- You may assume that all N weights are distinct and that N is at least 6

2. **Winning lottery ticket detector.** Every lottery ticket carries a number on it. The organisers do not announce a single winning number — instead a ticket wins if its number passes a test. A ticket is a **winning ticket** if the sum of only its **even** digits is greater than 10. Odd digits are ignored, and 0 counts as an even digit. [CO4] (8)

Write a C program that takes the ticket number from the user and prints:

- every even digit found in the ticket number
- the sum of those even digits
- whether the ticket is a winning ticket or not

You must work the digits out using arithmetic only. Do not turn the number into a string and do not use any built-in function.

Sample: ticket 68234 has even digits 6, 8, 2 and 4, which add up to 20. As 20 is greater than 10 it **is** a winning ticket. Ticket 4321 has even digits 4 and 2, adding up to only 6, so it is **not**.

3. The following program is supposed to read the marks of **N students** into an array, calculate the **average** mark, and then count and print how many students scored **above the average**. The program does not work. Identify every error and write the corrected line. [CO4] (7)

No.	Code	Explanation/Corrected
1		
2	int main() {	
3	int n, i, above;	
4	float marks[100], sum, avg;	
5	printf("Enter number of students: ");	
6	scanf("%d", n);	
7	for(i=1; i<=n; i++) {	
8	scanf("%f", &marks[i]);	
9	sum = sum + marks[i];	
10	}	
11	avg = sum / N;	
12	printf("Average = %d\n", avg);	
13	above = 0;	
14	for(i=0; i<n; i++) {	
15	if (marks[i] => avg)	
16	above = above + 1;	
17	}	
18	printf("Above average: %d\n" above);	
19	}	